

AI Teaching: Principles, Methods, and Classroom Practice

A practical reference document for an AI Teacher and Retrieval-Augmented Generation (RAG) evaluation project.

1. Introduction to AI in Education

What is Artificial Intelligence?

Artificial Intelligence (AI) refers to computer systems that can perform tasks that normally require human intelligence. These tasks include understanding language, recognizing patterns, generating text, solving problems, making predictions, and assisting with decisions. In education, AI can support teachers and students by providing explanations, generating learning materials, giving feedback, and adapting activities to different learning needs.

AI Teaching

AI teaching means using artificial intelligence to support the teaching and learning process. AI should be treated as a teaching aid rather than a complete replacement for teachers. Human teachers provide context, judgment, empathy, classroom management, and educational guidance. AI can handle repetitive tasks and provide additional learning support.

2. Role of AI in Teaching and Learning

Personalized Learning

AI can personalize learning by adjusting explanations, examples, questions, and practice activities according to a student's needs. A student who is struggling with a concept can receive simpler explanations and additional examples, while an advanced student can receive more challenging questions.

Intelligent Tutoring

An AI tutor can answer questions, explain concepts step by step, provide hints, and generate practice exercises. A useful AI tutor should encourage students to understand the reasoning behind an answer instead of simply providing the final answer.

Teacher Assistance

Teachers can use AI to prepare lesson plans, create quizzes, generate discussion questions, summarize educational material, produce differentiated activities, and draft feedback. The teacher should review AI-generated material before using it with students.

Student Engagement

AI can support interactive learning through simulations, question-and-answer activities, role-play, brainstorming, and personalized practice. Well-designed AI activities can encourage students to ask questions and explore alternative solutions.

3. Generative AI and Large Language Models

Generative AI

Generative AI is a type of artificial intelligence that can create new content based on instructions. It can generate text, explanations, questions, examples, summaries, images, code, and other forms of content.

Large Language Models

Large Language Models (LLMs) are AI systems trained on large collections of text. They can generate human-like responses to prompts. Educational uses include explaining difficult topics, creating practice questions, rewriting material at different reading levels, and helping students brainstorm ideas.

Limitations

Generative AI can produce incorrect information, incomplete explanations, biased content, or fabricated references. This is sometimes called hallucination. Students and teachers should verify important information using reliable sources.

4. Effective Prompting for Teachers

What is a Prompt?

A prompt is an instruction or question given to an AI system. The quality of the prompt strongly affects the quality of the response.

Prompt Structure

A useful educational prompt can specify the role of the AI, the task, the target audience, the required format, relevant context, and constraints.

Example Prompt

Act as a secondary-school science teacher. Explain photosynthesis in simple language for a student who has difficulty understanding biology. Use one everyday analogy, list the key steps, and provide three short practice questions.

5. Retrieval-Augmented Generation (RAG)

What is RAG?

Retrieval-Augmented Generation, or RAG, is an approach in which an AI model retrieves relevant information from a collection of documents before generating an answer. This helps the model answer questions using a specified knowledge source.

RAG Process

A typical RAG pipeline loads documents, divides them into smaller chunks, converts each chunk into an embedding, stores embeddings in a vector database such as FAISS, retrieves relevant chunks for a question, and provides those chunks to a language model as context.

Educational Use of RAG

RAG can be used to build AI tutors that answer questions from a course textbook, school policy document, lecture notes, or teaching handbook. An AI teacher can be instructed to answer only from a supplied teaching document and state that information could not be found when the document does not contain the answer.

6. AI Teacher Workflow

Step 1: Understand the Question

The AI system identifies what the learner is asking and determines the required type of response.

Step 2: Retrieve Relevant Information

For a RAG-based teacher, the system searches the knowledge base and retrieves the most relevant document chunks.

Step 3: Generate an Answer

The language model uses the question and retrieved context to produce an answer. The prompt can instruct the model to stay within the retrieved information.

Step 4: Evaluate the Answer

The generated answer can be evaluated for correctness, relevance, faithfulness to the source, completeness, and clarity.

Step 5: Provide Feedback

The AI teacher can provide constructive feedback, explain mistakes, and suggest additional practice.

7. Evaluation of AI Teacher Answers

Reference Answer

A reference answer is a trusted answer used as a comparison point when evaluating an AI-generated response. It represents the expected or ideal answer for a question.

Answer Relevance

Answer relevance measures whether the AI response directly addresses the user's question. An answer can be factually correct but still have low relevance if it does not answer what was asked.

Faithfulness

Faithfulness measures whether an answer is supported by the retrieved context. A faithful RAG answer should not introduce unsupported claims.

Contextual Precision

Contextual precision measures whether the retrieved information is relevant to the question. High precision means the retrieved chunks contain useful information rather than unrelated material.

Contextual Recall

Contextual recall measures whether the retrieval process found the important information needed to answer the question.

Correctness

Answer correctness compares an AI-generated response with a trusted reference answer or reliable evidence.

8. Operational Metrics

Latency

Latency is the amount of time required to produce an AI response. Lower latency generally provides a better interactive learning experience.

Token Usage

Tokens are units of text processed by a language model. Token counts can be used to measure input and output processing.

Cost

Cloud AI services may charge according to input and output token usage. Local models can avoid per-token API charges, although they require computing resources.

Reliability

Reliability describes whether an AI system consistently produces responses without errors, timeouts, or unavailable-service failures.

9. Benefits of AI in Education

Accessibility

AI can provide explanations in different styles and levels of difficulty and support learners who need repeated explanations or additional practice.

Efficiency

AI can reduce the time teachers spend on repetitive preparation tasks such as drafting quizzes, generating examples, and organizing learning material.

Personalization

AI can adapt content to individual learners by changing difficulty, examples, language complexity, and feedback.

Continuous Support

An AI tutor can provide assistance outside normal classroom hours.

10. Risks and Challenges

Incorrect Information

AI systems can generate incorrect answers with high confidence. Teachers should verify important content and teach students to check sources.

Bias

AI outputs can reflect biases present in training data or prompts. Educational use should include awareness of fairness and representation.

Privacy

Schools should avoid sending sensitive student information to AI systems unless appropriate privacy protections are in place.

Overdependence

Students may become dependent on AI if they use it to complete tasks without learning the underlying concepts. AI should support learning rather than replace thinking.

Academic Integrity

Students may use generative AI to produce work that they submit as their own. Schools should establish clear guidelines for acceptable AI use.

11. Best Practices for an AI Teacher

Use AI as a Support Tool

The teacher remains responsible for educational decisions. AI should support instruction rather than replace professional judgment.

Ground Answers in Trusted Sources

Use reliable textbooks, lecture notes, institutional documents, and other approved sources. RAG can help restrict answers to a defined knowledge base.

Ask AI to Explain

When appropriate, encourage explanations, examples, hints, and step-by-step learning rather than only final answers.

Verify Important Claims

Important facts, references, calculations, and recommendations should be checked against reliable sources.

Give Students Clear AI Rules

Students should know when AI is allowed, what assistance is acceptable, and when they must complete work independently.

12. Example AI Teaching Scenario

Scenario

A teacher provides an AI system with a course PDF about artificial intelligence. A student asks, 'What is retrieval-augmented generation?' The RAG system searches the course document, retrieves the sections describing RAG, and provides those sections to the language model.

Expected Behavior

The AI teacher should explain RAG using the retrieved material. If the student asks for information that is not present in the document, the AI should clearly state that the information could not be found in the provided document rather than inventing an answer.

13. Key Terms

Artificial Intelligence

Computer systems designed to perform tasks associated with human intelligence.

Generative AI

AI that creates new content such as text, images, or code.

LLM

Large Language Model capable of processing and generating natural language.

Prompt

An instruction or input provided to an AI system.

Embedding

A numerical representation of text used to measure semantic similarity.

Vector Database

A system used to store and search numerical vector representations.

FAISS

A library for efficient similarity search over vectors.

RAG

Retrieval-Augmented Generation, which combines information retrieval with text generation.

Retriever

A component that finds relevant documents or chunks for a query.

Hallucination

An AI-generated statement that is unsupported or factually incorrect.

Faithfulness

The degree to which an answer is supported by the supplied context.

Latency

The time taken by a system to produce a response.